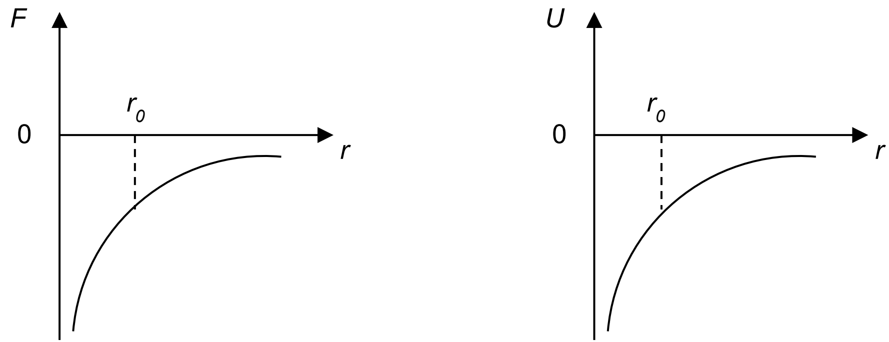
10 A small point mass is in a gravitational field setup by a massive perfect sphere of uniform density. The graphs below show the variation of gravitational force F on the small point mass and gravitational potential energy U with distance between the centres of both masses r. Both graphs are not drawn to scale.



When the two masses are separated by a distance r_0, which of the following is correct?

A The U of two masses is represented by the gradient of the tangent at r_0 in the F-r graph.

B The magnitude of F on the small point mass is represented by the ratio of U to r_0 in the U-r graph.

C The U of two masses is represented by the area under the F-r graph from 0 to r_0.

D The magnitude of F on the massive perfect sphere is represented by the gradient of the tangent at r_0 in the U-r graph.